



2017 HSC Course Assessment Task 3 (Trial Examination)

Friday 23rd of June, 2017

- Working time – 3 hours.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.

- Mark your answers on the answer sheet provided (numbered as page 13)

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

Class (please ✓)

○ 12M4C – Mr Lin

Marker's use only.

[illegible]

Section I

10 marks

Attempt Question 1 to 10

Allow approximately 15 minutes for this section

Mark your answers on the answer sheet provided.

Questions

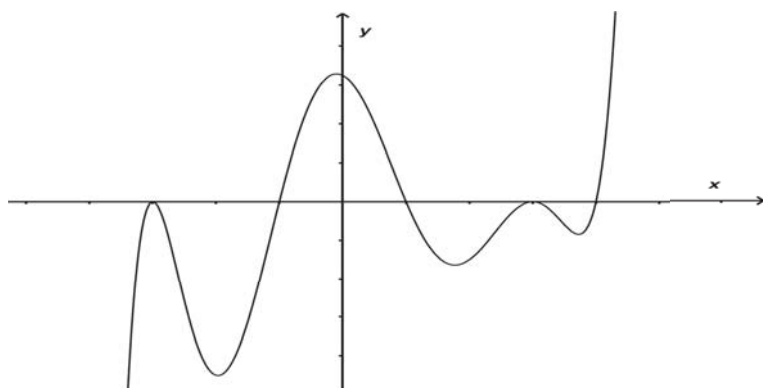
Marks

1. Let $z = 5 - i$ and $\omega = 2 + 3i$. What is the value of $2z + \bar{\omega}$? 1
- (A) $12 + i$
- (B) $12 + 2i$
- (C) $12 - 4i$
- (D) $12 - 5i$
2. If $-2 + 2i\sqrt{3}$ is expressed in modulus-argument form, the result is 1
- (A) $4\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$
- (B) $2\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$
- (C) $2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$
- (D) $4\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$
3. It is known that $2 + 3i$ is a solution to $x^4 - 6x^3 + 26x^2 - 46x + 65 = 0$. Another solution is 1
- (A) $-2 - 3i$
- (B) $-1 - 2i$
- (C) $1 - 2i$
- (D) $-2 + i$

4. Let α , β , and γ be the roots of the equation $x^3 + px^2 + q = 0$. The polynomial with roots 2α , 2β and 2γ is: 1
- (A) $x^3 - 2px^2 + 8q = 0$
- (B) $x^3 + 2px^2 + 4q = 0$
- (C) $x^3 - 2px^2 - 8q = 0$
- (D) $x^3 + 2px^2 + 8q = 0$
5. Given the eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is e , then the eccentricity of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is 1
- (A) $-e$
- (B) $\frac{1}{e}$
- (C) e
- (D) e^2
6. $\int \frac{1}{1 + \operatorname{cosec} x} dx =$ 1
- (A) $\frac{\left(1 + \tan \frac{x}{2}\right)^2}{1 + \left(\tan \frac{x}{2}\right)^2} + x + c$
- (B) $\frac{\left(1 - \tan \frac{x}{2}\right)^2}{1 + \left(\tan \frac{x}{2}\right)^2} - x + c$
- (C) $\sec x - \tan x + x + c$
- (D) $\tan x - \sec x - x + c$

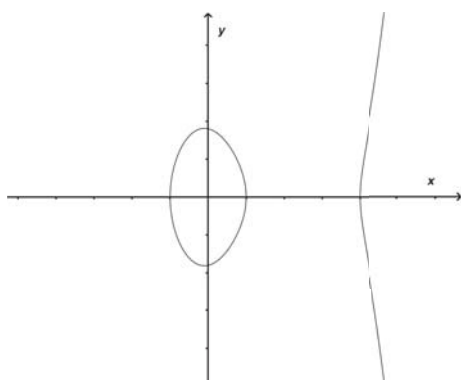
7. Consider the graph of $y = f(x)$ drawn below.

1

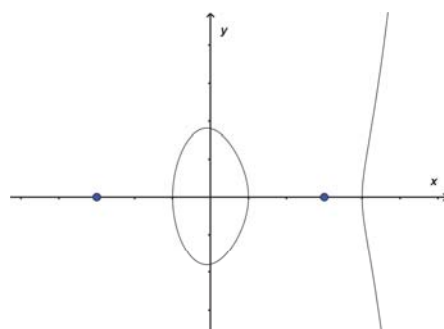


Which of the following diagrams show the graph of $y^2 = f(x)$?

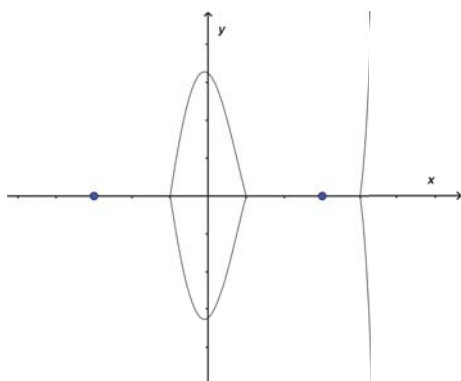
(A)



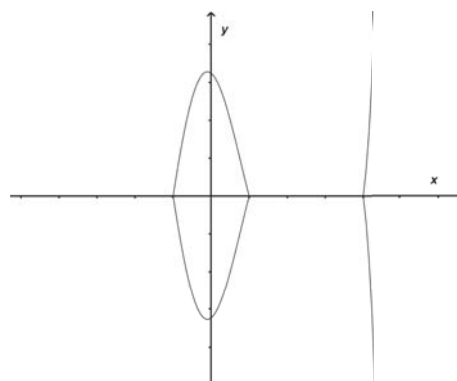
(C)



(B)



(D)



8. If $\sqrt{yx^2 + xy^2} = 3$, then at the point $(1, -1)$, the value of $\frac{dy}{dx}$ is 1

(A) 1

(B) -1

(C) $\frac{1}{3}$

(D) $-\frac{1}{3}$

9. The cross section perpendicular to the x -axis between two curves $y = \sqrt{x}$ and $y = 2\sqrt{x}$ is a circle. If the two curves are drawn between $x = 0$ and $x = 4$, the volume of the horn is given by 1

(A) $\int_0^4 \sqrt{x} \, dx$

(B) $\int_0^4 \pi \sqrt{x} \, dx$

(C) $\int_0^4 \frac{\pi}{2} x \, dx$

(D) $\int_0^4 \frac{\pi x}{4} \, dx$

10. The value of 1

$$\lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\frac{\pi}{2}-h}^{\frac{\pi}{2}+h} \frac{\sin x}{x} \, dx \right)$$

is

(A) 0

(B) 1

(C) $\frac{2}{\pi}$

(D) $\frac{4}{\pi}$

Examination continues overleaf...

Section II

90 marks

Attempt Questions 11 to 16

Allow approximately 2 hours and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks)	Commence a NEW page.	Marks
(a) Let $z = 4 + i$ and $w = \bar{z}$. Find $\frac{z}{w}$ in the form $x + iy$.		1
(b) Find $\int \frac{dx}{\sqrt{2x - x^2}}$.		2
(c) Given that $\frac{25}{(x-1)^2(x^2+4)} = \frac{ax+b}{(x-1)^2} + \frac{cx+d}{x^2+4}.$ i. Find a, b, c and d.		2
ii. Hence, find $\int \frac{25}{(x-1)^2(x^2+4)} dx$.		3
(d) The equation $z^5 = 1$ has roots $1, \omega, \omega^2, \omega^3, \omega^4$, where $\omega = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$.		
i. Show that $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$.		1
ii. Show that $\left(\omega + \frac{1}{\omega}\right)^2 + \left(\omega + \frac{1}{\omega}\right) - 1 = 0$.		1
iii. Hence, show that $\cos \frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$.		2
(e) The region enclosed by the curves $y = x + 1$ and $y = (x - 1)^2$ is rotated about the y -axis. Find the volume of the solid formed.		3

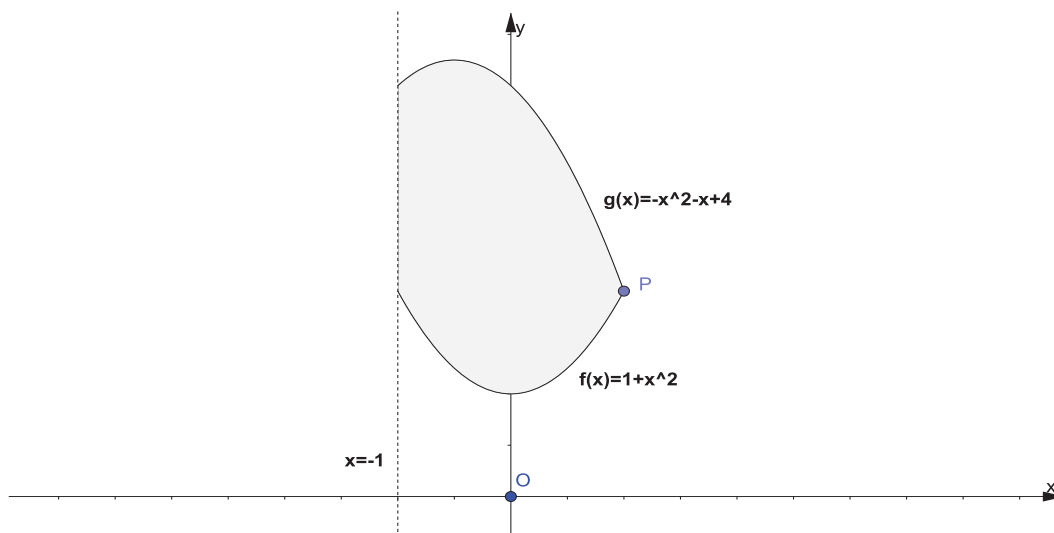
End of Question 11

Question 12 (15 Marks)

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Marks

- (a) For the ellipse $\frac{x^2}{16} + \frac{y^2}{25} = 1$, find:
- The eccentricity. **1**
 - The coordinates of the foci S and S' and the equations of its directrices. **2**
 - Sketch the ellipse showing all the above features. **1**
- (b) Given the polynomial $P(x) = 2x^3 + 3x^2 - x + 1$ has roots α, β and γ :
- Find the polynomial whose roots are α^2, β^2 and γ^2 . **2**
 - Determine the value of $\alpha^3 + \beta^3 + \gamma^3$. **3**
- (c) The shaded region bounded by $g(x) = -x^2 - x + 4$, $f(x) = 1 + x^2$ and $x = -1$ is rotated about the line $x = -1$. The point P is the intersection of $f(x)$ and $g(x)$ in the first quadrant.



- Find the x -coordinate of P . **1**
- Use the method of cylindrical shells to express the volume of the resulting solid of revolution as an integral. **3**
- Evaluate the integral in part (ii). **2**

End of Question 12

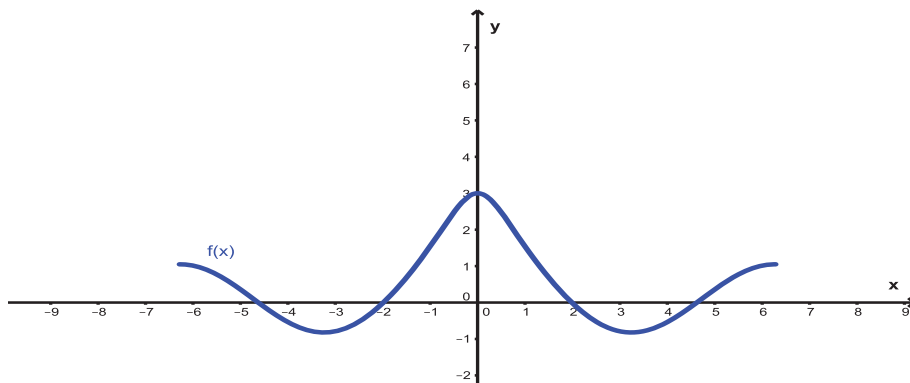
Question 13 (15 Marks)

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Marks

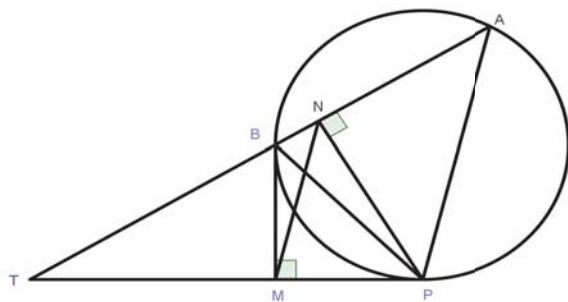
- (a) Use integration by parts to evaluate $\int x \ln(x^3 + x) dx$. **3**

- (b) The diagram shows the graph of the function $y = f(x)$.



Draw separate one-third page sketches of graphs of the following:

- i. $y = \sqrt{f(x)}$ **2**
 - ii. $|y| = f(x)$ **2**
 - iii. $y = f(x)^2$ **2**
 - iv. $y = e^{-f(x)}$ **2**
- (c) The points A, B and P lie on a circle. The chord AB produced and the tangent at P intersect at the point T , as shown in the diagram. The point N is the foot of the perpendicular to AB through P , and the point M is the foot of the perpendicular to PT through B .



Copy or trace this diagram into your writing booklet.

- i. Explain why $BNPM$ is a cyclic quadrilateral. **1**
- ii. Prove that MN is parallel to PA . **3**

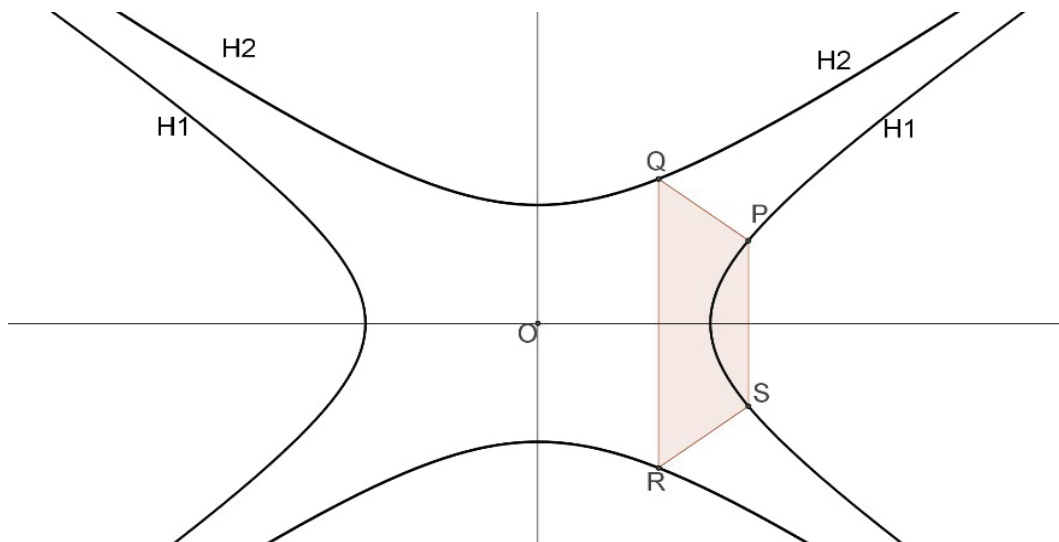
End of Question 13

Question 14 (15 Marks)

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Marks

- (a) The hyperbola $\mathcal{H}_1 : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $\mathcal{H}_2 : \frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$ as shown in the diagram. The line $x = a \sec \theta$ cuts $\mathcal{H}_1$ at P and S . Similarly the line $x = a \tan \theta$ cuts $\mathcal{H}_2$ at Q and R .



- i. Show that the y -coordinates of P and S are $\pm b \tan \theta$ respectively and the y -coordinates of Q and R are $\pm b \sec \theta$ respectively. **2**
 - ii. Prove that the area of trapezium $PQRS$ is independent of θ . **2**
 - iii. Show that the equation of the line PQ is $bx + ay = ab(\tan \theta + \sec \theta)$. **2**
 - iv. Prove that the area of triangle OPQ equals to half the area of the trapezium $PQRS$. **3**
- (b) Given that $I_n = \int_0^1 (1 - x^2)^n dx$.
- i. Evaluate I_1 and I_2 . **2**
 - ii. Show that $I_{n+1} = \frac{2(n+1)}{2n+3} I_n$. **2**
 - iii. Hence or otherwise prove that $I_n = \frac{2^{2n}(n!)^2}{(2n+1)!}$. **2**

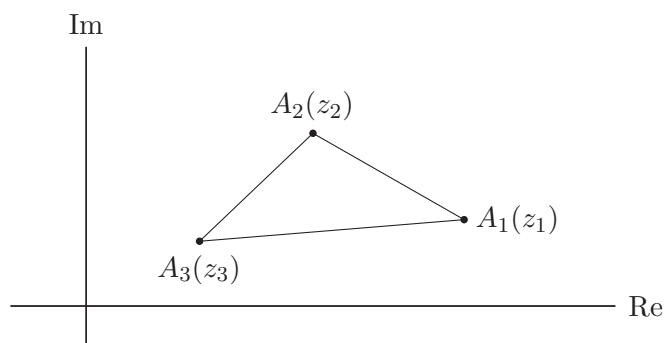
End of Question 14

Question 15 (15 Marks)

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Marks

- (a) $A_1A_2A_3$ is an equilateral triangle, the vertices occurring in the positive direction of rotation.



- i. Prove by geometric means or otherwise that

2

$$\overrightarrow{A_3A_1} = \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right) \overrightarrow{A_2A_3} = \omega \overrightarrow{A_2A_3},$$

where ω is a complex cube root of unity.

- ii. z_1 , z_2 and z_3 are the complex numbers corresponding to A_1 , A_2 and A_3 respectively. The triangle $A_1A_2A_3$ is inscribed in a circle of centre z_0 and radius r . Show that $z_0 = \frac{1}{3}[z_1 + z_2 + z_3]$ and $r = \frac{1}{\sqrt{3}}|z_1 - z_2|$.

3

- iii.

Use (i) or otherwise, prove that $z_1 + \omega z_2 + \omega^2 z_3 = 0$.

2

- iv. Hence or otherwise, prove that

3

$$z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1.$$

- (b) Given that $x_n = \frac{1}{2} \left[(1 + i\sqrt{2})^n + (1 - i\sqrt{2})^n \right]$, $n \geq 0$.

Let $y_0 = 6$, $y_1 = 2$ and $3y_n = 2y_{n-1} - y_{n-2}$, $n \geq 2$.

- i. Prove by mathematical induction that $y_n = \frac{2}{3^{n-1}}x_n$, $n \geq 0$.

3

- ii. Hence or otherwise, show that

2

$$y_n = 3 \left[\left(\frac{1}{1 + i\sqrt{2}} \right)^n + \left(\frac{1}{1 - i\sqrt{2}} \right)^n \right], n \geq 0.$$

End of Question 15

Question 16 (15 Marks)

Commence a NEW page.

Marks

- (a) i. Given that $a + b \geq 2\sqrt{ab}$. Prove that $a^2 + b^2 + c^2 \geq ab + ac + bc$. **1**
- ii. Given that $a + b + c \geq 3\sqrt[3]{abc}$. Hence or otherwise prove that **3**
- $$\frac{a^3}{b-c} + \frac{b^3}{c-a} + \frac{c^3}{a-b} \geq \frac{3}{2}(ab + ac + bc).$$
- (b) i. Prove that for all positive values of x , $x > \ln(1+x)$. **3**
- ii. Given that $x_n = (1 + \frac{1}{3})(1 + \frac{1}{3^2}) \dots (1 + \frac{1}{3^n})$, $n \geq 1$. Show that $x_{n+1} > x_n$. **1**
- iii. Prove that **3**
- $$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{x_{k+1} - x_k}{x_k} = \frac{1}{6}.$$
- iv. Show that $\ln x_n < \sum_{k=1}^n \frac{1}{3^k}$. **2**
- v. Hence or otherwise, show that $x_n < \sqrt{e}$ for all positive integer n . **2**

End of paper.

MC:

1. $z = 5 - i, \omega = 2 + 3i$

$$2z + \bar{\omega} = 2(5 - i) + 2 - 3i = 12 - 5i$$

The answer is D.

2. $-2 + 2i\sqrt{3} = 4\left(\frac{-1}{2} + i\frac{\sqrt{3}}{2}\right) = 4\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$

The answer is D.

3. $2 + 3i$ is a root of $x^4 - 6x^3 + 26x^2 - 46x + 65 = 0$
 $2 - 3i$ is also a root (complex conjugate theorem).

Assume $a + ib$ is another root, so $a - ib$.

Sum of roots are $= 2 + 3i + 2 - 3i + a + ib + a - ib = 4 + 2a$

But sum of roots = 6. Hence $a=1$. And the answer is C.

4. Replace x by $\frac{x}{2}$. $\left(\frac{x}{2}\right)^3 + p\left(\frac{x}{2}\right)^2 + q = 0$

$$\frac{x^3}{8} + p\frac{x^2}{4} + q = 0$$

$$x^3 + 2px^2 + 8q = 0$$

The answer is D.

5. $b^2 = a^2(e^2 - 1) \therefore e^2 = \frac{a^2 + b^2}{a^2}$

Let E the eccentricity of the ellipse $\therefore b^2 = (a^2 + b^2)(1 - E^2)$

$$E^2 = \frac{a^2}{a^2 + b^2} = \frac{1}{e^2} \therefore E = \frac{1}{e}$$

The answer is B.

6. $\int \frac{1}{1 + \csc x} dx = \int \frac{\sin x}{1 + \sin x} dx = \int \frac{\sin x}{1 + \sin x} \times \frac{1 - \sin x}{1 - \sin x} dx = \int \frac{\sin x - (\sin x)^2}{(\cos x)^2} dx =$

$$\int \frac{\sin x}{(\cos x)^2} dx - \int (\tan x)^2 dx = - \int \frac{du}{u^2} - \int ((\sec x)^2 - 1) dx$$

$$\frac{1}{u} - \tan x + x + c = \frac{1}{\cos x} - \tan x + x + c = \sec x - \tan x + x + c$$

(for $\int \frac{\sin x}{(\cos x)^2} dx$, use $u = \cos x$)

The answer is C.

7. The answer is C.

8. $\sqrt{yx^2 + xy^2} = 3 \therefore yx^2 + xy^2 = 9$

$$2xy + x^2 \frac{dy}{dx} + y^2 + 2xy \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{2xy + y^2}{x^2 + 2xy}$$

At $(1, -1)$, $\frac{dy}{dx} = -\frac{2 \times 1 \times -1 + (-1)^2}{1^2 + 2 \times 1 \times -1} = -\frac{-1}{-1} = -1$

The answer is B.

9. The diameter of the circle is $2\sqrt{x} - \sqrt{x} = \sqrt{x}$

So the area of the circle is $\pi \left(\frac{d}{2}\right)^2 = \pi \frac{x}{4}$

Volume = $\int_0^4 \pi \frac{x}{4} dx$. The answer is D.

$$\begin{aligned}
10. \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\frac{\pi}{2}-h}^{\frac{\pi}{2}+h} \frac{\sin x}{x} dx \right) &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\frac{\pi}{2}-h}^{\frac{\pi}{2}} \frac{\sin x}{x} dx \right) + \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\frac{\pi}{2}}^{\frac{\pi}{2}+h} \frac{\sin x}{x} dx \right) \\
&= \lim_{h \rightarrow 0} \frac{1}{h} \left[F\left(\frac{\pi}{2}\right) - F\left(\frac{\pi}{2}-h\right) \right] + \lim_{h \rightarrow 0} \frac{1}{h} \left[F\left(\frac{\pi}{2}+h\right) - F\left(\frac{\pi}{2}\right) \right] \\
&= \frac{\sin \frac{\pi}{2}}{\frac{\pi}{2}} + \frac{\sin \frac{\pi}{2}}{\frac{\pi}{2}} = \frac{4}{\pi}
\end{aligned}$$

Where $\frac{dF}{dx} = \frac{\sin x}{x}$.

The answer is D.

Question 11.

a) $z = 4 + i, \omega = \bar{z} = 4 - i$

$$\frac{z}{\omega} = \frac{4+i}{4-i} = \frac{4+i}{4-i} \times \frac{4+i}{4+i} = \frac{16+8i-1}{16+1} = \frac{15}{17} + \frac{8}{17}i$$

b) $\int \frac{1}{\sqrt{2x-x^2}} dx = \int \frac{dx}{\sqrt{1-1+2x-x^2}} = \int \frac{dx}{\sqrt{1-(x-1)^2}} = \int \frac{du}{\sqrt{1-u^2}} = \sin^{-1} u + c = \sin^{-1}(x-1) + c$
(where $u = x-1$).

c) i) $\frac{25}{(x-1)^2(x^2+4)} = \frac{ax+b}{(x-1)^2} + \frac{cx+d}{x^2+4}$
 $\therefore 25 = (ax+b)(x^2+4) + (cx+d)(x-1)^2$
 $25 = ax^3 + 4ax + bx^2 + 4b + (cx+d)(x^2-2x+1)$
 $25 = ax^3 + 4ax + bx^2 + 4b + cx^3 - 2cx^2 + cx + dx^2 - 2dx + d$
 $25 = (a+c)x^3 + (b-2c+d)x^2 + (4a+c-2d)x + 4b+d$
 $\therefore a+c=0$ (1), $b-2c+d=0$ (2), $4a+c-2d=0$ (3) and $4b+d=25$ (4)
(1) $\rightarrow c = -a$

(1)&(3) $\rightarrow 3a-2d=0$, or $d = \frac{3}{2}a$ (5)

(1)&(2) $\rightarrow b+d = -2a$, or $b = -\frac{7}{2}a$ (6)

(4), (5) and (6) $\rightarrow -14a + \frac{3}{2}a = 25 \rightarrow a = -2$

So $b = 7, c = 2$ and $d = -3$

ii) $\int \frac{25}{(x-1)^2(x^2+4)} dx = \int \frac{-2x+7}{(x-1)^2} dx + \int \frac{2x-3}{x^2+4} dx =$
 $-2 \int \frac{x-1}{(x-1)^2} dx + 5 \int \frac{1}{(x-1)^2} dx + \int \frac{2x}{x^2+4} dx - 3 \int \frac{1}{x^2+4} dx$
 $= -2 \ln|x-1| - \frac{5}{x-1} + \ln(x^2+4) - \frac{3}{2} \tan^{-1} \frac{x}{2} + c$

d) i) $z^5 = 1 \rightarrow z^5 - 1 = 0 \rightarrow (z-1)(1+z+z^2+z^3+z^4) = 0 \rightarrow 1+z+z^2+z^3+z^4 = 0$

since ω is a root of $z^5 - 1 = 0$. $\rightarrow 1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$.

ii) $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$. Divide by ω^2 , we obtain:

$$\frac{1}{\omega^2} + \frac{1}{\omega} + 1 + \omega + \omega^2 = 0$$

$$\left(\frac{1}{\omega^2} + \omega^2 \right) + \left(\frac{1}{\omega} + \omega \right) + 1 = 0$$

$$\left(\frac{1}{\omega^2} + \omega^2 + 2\right) + \left(\frac{1}{\omega} + \omega\right) + 1 - 2 = 0$$

$$\left(\frac{1}{\omega} + \omega\right)^2 + \left(\frac{1}{\omega} + \omega\right) - 1 = 0 (***)$$

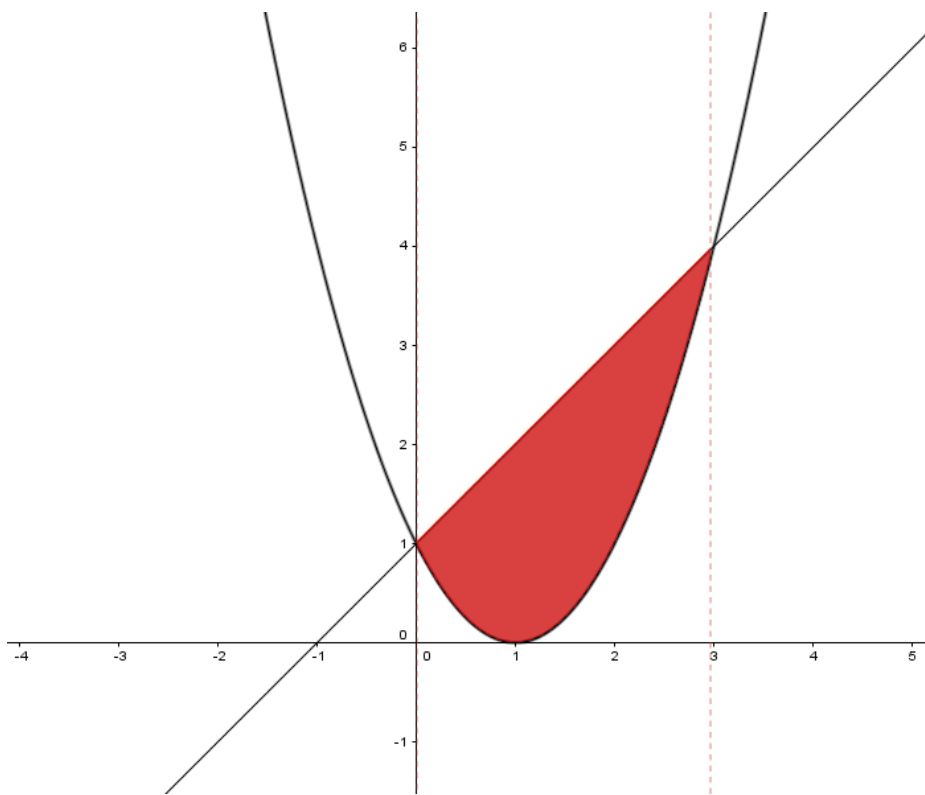
$$\text{iii) But } \omega = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}, \text{ and } \frac{1}{\omega} = \bar{\omega}, \text{ so } \frac{1}{\omega} + \omega = 2 \cos \frac{2\pi}{5}.$$

$$\text{Let } X = \frac{1}{\omega} + \omega, \text{ so } (***) \rightarrow X^2 + X - 1 = 0 \text{ and } X = \frac{-1 \pm \sqrt{(1)^2 - 4(1)(-1)}}{2}$$

$$\text{Since } \frac{2\pi}{5} \text{ is in the first quadrant, the cosine will be positive. } 2 \cos \frac{2\pi}{5} = \frac{-1 + \sqrt{5}}{2}$$

$$\cos \frac{2\pi}{5} = \frac{-1 + \sqrt{5}}{4}.$$

e)



Using the cylindrical Shell Method.

$$x + 1 = (x - 1)^2 = x^2 - 2x + 1 \therefore x^2 - 3x = 0 \therefore x = 0 \text{ or } x = 3.$$

$$\delta V = 2\pi r h \delta x = 2\pi x(x + 1 - (x - 1)^2) \delta x = 2\pi x(3x - x^2) \delta x$$

$$V = 2\pi \lim_{\delta x \rightarrow 0} \sum_{x=0}^{x=3} (3x^2 - x^3) \delta x$$

$$V = 2\pi \int_0^3 (3x^2 - x^3) dx = 2\pi \left[x^3 - \frac{x^4}{4} \right]_0^3 = 2\pi \left(27 - \frac{81}{4} - 0 \right) = \frac{27}{2} \pi U^3.$$

Using the Washer method:

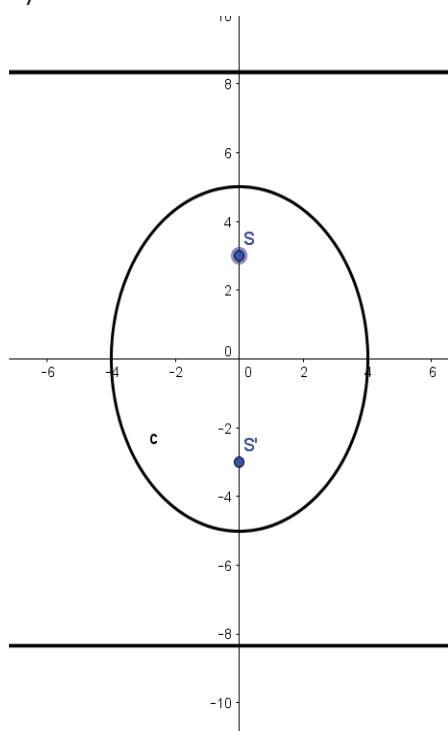
$$V = \pi \int_0^1 [((1 + \sqrt{y})^2 - (1 - \sqrt{y})^2)] dx + \pi \int_1^4 [((1 + \sqrt{y})^2 - (y - 1)^2)] dx$$

For the first integral needs to solve for x : $y = x^2 - 2x + 1, x = 1 \pm \sqrt{y}$.

$$V = \frac{65\pi}{6} + \frac{8\pi}{6} = \frac{27}{2}\pi U^3.$$

Question 12

- a) i) $16 = 25(1 - e^2) \therefore e^2 = \frac{9}{25}, e = \frac{3}{5}$.
 ii) $S(0,3), S'(0,-3)$, directrices $y = \pm \frac{b}{e} = \pm \frac{25}{3}$
 iii)



- b) $P(x) = 2x^3 + 3x^2 - x + 1$
 i) Replace x by $\sqrt{x}$ in $P(x) = 0$.

$$2(\sqrt{x})^3 + 3(\sqrt{x})^2 - \sqrt{x} + 1 = 0 \therefore 2x\sqrt{x} + 3x - \sqrt{x} + 1 = 0$$

$$\sqrt{x}(2x - 1) + 3x + 1 = 0 \therefore 3x + 1 = \sqrt{x}(1 - 2x)$$

$$(3x + 1)^2 = (\sqrt{x}(1 - 2x))^2 \therefore 9x^2 + 6x + 1 = x(1 - 4x + 4x^2)$$

$$4x^3 - 13x^2 - 5x - 1 = 0.$$

 ii) α, β, γ are roots of $2x^3 + 3x^2 - x + 1 = 0$

$$\alpha + \beta + \gamma = -\frac{3}{2}$$

 $\alpha^2, \beta^2, \gamma^2$ are roots of $4x^3 - 13x^2 - 5x - 1 = 0$.

$$\alpha^2 + \beta^2 + \gamma^2 = \frac{13}{4}$$

α, β, γ are roots of $2x^3 + 3x^2 - x + 1 = 0$

$$2\alpha^3 = -3\alpha^2 + \alpha - 1$$

$$2\beta^3 = -3\beta^2 + \beta - 1$$

$$2\gamma^3 = -3\gamma^2 + \gamma - 1$$

$$\therefore 2(\alpha^3 + \beta^3 + \gamma^3) = -3(\alpha^2 + \beta^2 + \gamma^2) + \alpha + \beta + \gamma - 3 = -3\frac{13}{4} + \frac{-3}{2} - 3 = \frac{-57}{4}$$

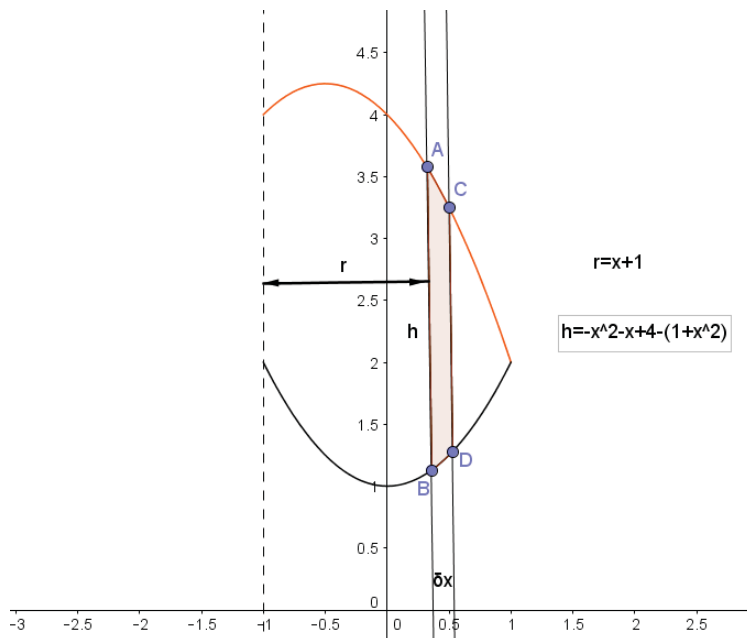
$$\alpha^3 + \beta^3 + \gamma^3 = \frac{-57}{8}$$

c)

i) $-x^2 - x - 4 = 1 + x^2 \therefore 2x^2 + x - 3 = 0$

$$(2x + 3)(x - 1) = 0 \therefore x = \frac{-3}{2} \text{ and } x = 1.$$

P is in the first quadrant $\therefore x = 1$.



ii) $\delta V = 2\pi r h \delta x =$

$$2\pi(x + 1)(-x^2 - x + 4 - 1 - x^2)\delta x = 2\pi(x + 1)(3 - x - x^2)\delta x$$

$$V = 2\pi \lim_{\delta x \rightarrow 0} \sum_{x=-1}^{x=1} (x + 1)(3 - x - 2x^2)\delta x$$

$$V = 2\pi \int_{-1}^1 (x + 1)(3 - x - 2x^2)dx$$

iii) $V = 2\pi \int_{-1}^1 (x + 1)(3 - x - 2x^2)dx = 2\pi \int_{-1}^1 (-2x^3 - 3x^2 + 2x + 3)dx$

$$= 2\pi \left[-\frac{x^4}{4} - x^3 + x^2 + 3x \right]_{-1}^1$$

$$2\pi \left[\frac{-1}{2} - 1 + 1 + 3 - \left(\frac{-1}{2} + 1 + 1 - 3 \right) \right] = 2\pi[4] = 8\pi U^3.$$

Question 13

a) $\int x \ln(x^3 + x) dx$ Let $u = \ln(x^3 + x)$ and $dv = x dx$

$$\frac{du}{dx} = \frac{3x^2+1}{x(x^2+1)} = \frac{3(x^2+1)}{x(x^2+1)} - \frac{2}{x(x^2+1)} \text{ and } v = \frac{x^2}{2}$$

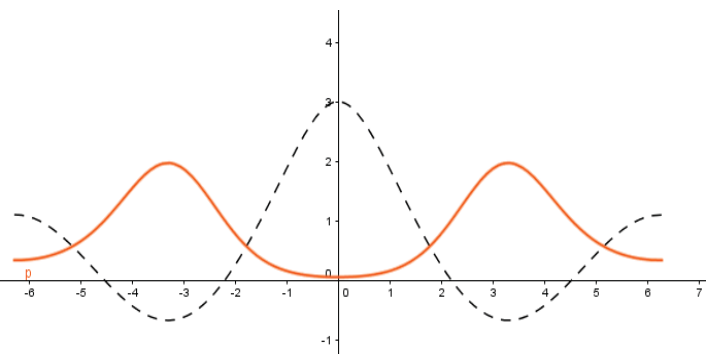
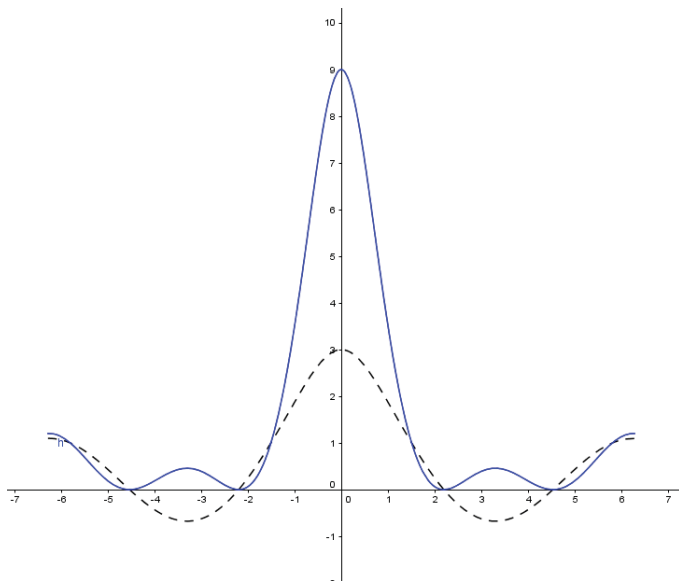
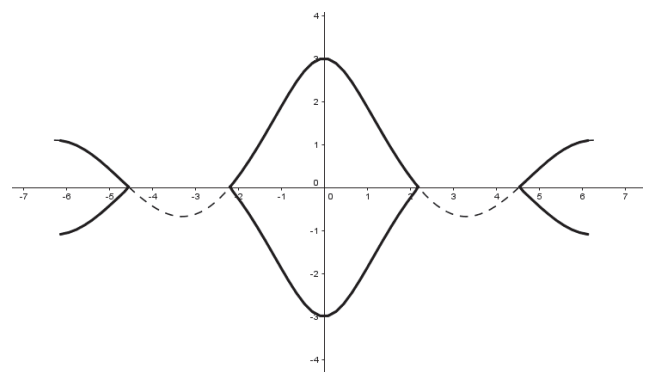
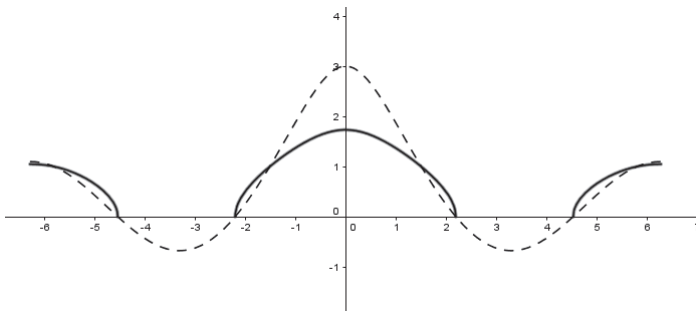
$$\int x \ln(x^3 + x) dx = \frac{x^2}{2} \ln(x^3 + x) - \frac{1}{2} \int x^2 \frac{3(x^2 + 1)}{x(x^2 + 1)} dx + \frac{1}{2} \int x^2 \frac{2}{x(x^2 + 1)} dx$$

$$= \frac{x^2}{2} \ln(x^3 + x) - \frac{3}{2} \int x dx + \frac{1}{2} \int \frac{2x}{x^2 + 1} dx$$

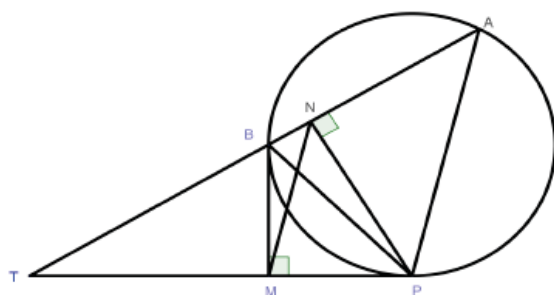
$$= \frac{x^2}{2} \ln(x^3 + x) - \frac{3}{2} \frac{x^2}{2} + \frac{1}{2} \ln(x^2 + 1) + c$$

$$= \frac{x^2}{2} \ln(x^3 + x) - \frac{3}{4} x^2 + \frac{1}{2} \ln(x^2 + 1) + c$$

b)



c)



- (i) In the quadrilateral BNPM, $\angle BMP + \angle BNP = 180^\circ$
(Opposite angles in cyclic quadrilateral are supplementary.)
- (ii) $\angle TPB = \angle TAP$ (angle between the tangent is equal to angle in the alternate segment = θ say). $\angle NPA = 90 - \theta$.
 $\angle NBP = 90 - \theta$. But $\angle MBP = \angle MNP$ (angles in the same segment).
 $\therefore \angle MNP = \angle NPA$.
But $\angle MNP$ and $\angle NPA$ are alternate angle on lines MN and PA. $\therefore MN \parallel PA$.
-

Question 14.

- a) i) $x = a \sec \theta \therefore \frac{a^2 (\sec \theta)^2}{a^2} - \frac{y^2}{b^2} = 1 \therefore \frac{y^2}{b^2} = (\sec \theta)^2 - 1 = (\tan \theta)^2$
 $y^2 = (b \tan \theta)^2 \therefore y = \pm b \tan \theta$.
 $x = a \tan \theta \therefore \frac{a^2 (\tan \theta)^2}{a^2} - \frac{y^2}{b^2} = -1 \therefore \frac{y^2}{b^2} = (\tan \theta)^2 + 1 = (\sec \theta)^2$
 $y^2 = (b \sec \theta)^2 \therefore y = \pm b \sec \theta$
- ii) Let M be the foot of the Perpendicular from P to QR.
Area of trapezium PQRS = $\frac{1}{2} (PM)(PS + QR)$
 $PM = a \sec \theta - a \tan \theta = a(\sec \theta - \tan \theta)$
 $PS = 2y(P) = 2b \tan \theta$
 $QR = 2y(Q) = 2b \sec \theta$
Area of PQRS = $\frac{1}{2} a(\sec \theta - \tan \theta) \times 2b(\sec \theta + \tan \theta) = ab((\sec \theta)^2 - (\tan \theta)^2) = ab$.
- iii) Gradient of PQ = $\frac{b \sec \theta - b \tan \theta}{a \tan \theta - a \sec \theta} = -\frac{b}{a}$
Equation of PQ : $y - b \tan \theta = -\frac{b}{a}(x - a \sec \theta)$
 $\therefore bx + ay = ab(\sec \theta + \tan \theta)$.
- iv) d=Perpendicular distance from O to PQ is $\frac{|b \times 0 + a \times 0 - ab(\sec \theta + \tan \theta)|}{\sqrt{a^2 + b^2}}$
 $PQ = \sqrt{(b \sec \theta - b \tan \theta)^2 + (a \tan \theta - a \sec \theta)^2}$
 $= \sqrt{b^2 (\sec \theta)^2 - 2b^2 \sec \theta \tan \theta + b^2 (\tan \theta)^2 + a^2 (\tan \theta)^2 - 2a^2 \sec \theta \tan \theta + a^2 (\sec \theta)^2}$

$$= \sqrt{(a^2 + b^2)((\sec \theta)^2 + (\tan \theta)^2) - 2(a^2 + b^2) \sec \theta \tan \theta}$$

$$= \sqrt{(a^2 + b^2)(\sec \theta - \tan \theta)^2} = (\sec \theta - \tan \theta) \sqrt{a^2 + b^2}$$

Area of OPQ = $\frac{1}{2} \times d \times PQ = \frac{1}{2} \frac{|-ab(\sec \theta + \tan \theta)|}{\sqrt{a^2 + b^2}} \times (\sec \theta - \tan \theta) \sqrt{a^2 + b^2} = \frac{1}{2} ab =$
 $\frac{1}{2}$ Area of PQRS.

b)

i) $I_1 = \int_0^1 (1 - x^2) dx = \left[x - \frac{x^3}{3} \right]_0^1 = 1 - \frac{1}{3} - 0 = \frac{2}{3}$

$$I_2 = \int_0^1 (1 - x^2)^2 dx = \int_0^1 (1 - 2x^2 + x^4) dx = \left[x - 2 \frac{x^3}{3} + \frac{x^5}{5} \right]_0^1 = 1 - \frac{2}{3} + \frac{1}{5} - 0 = \frac{8}{15}$$

ii) $I_{n+1} = \int_0^1 (1 - x^2)^{n+1} dx = \int_0^1 (1 - x^2)(1 - x^2)^n dx =$
 $\int_0^1 (1 - x^2)^n dx - \int_0^1 x^2 (1 - x^2)^n dx = I_n - \int_0^1 x^2 (1 - x^2)^n dx$

Now for $\int_0^1 x^2 (1 - x^2)^n dx$, let $u = x$ and $dv = x(1 - x^2)^n dx$, then

$$du = dx \text{ and } v = -\frac{1}{2} \times \frac{(1 - x^2)^{n+1}}{n + 1}$$

So $\int_0^1 x^2 (1 - x^2)^n dx = \left[-\frac{1}{2} \times x \frac{(1 - x^2)^{n+1}}{n + 1} \right]_0^1 + \frac{1}{2(n+1)} \int_0^1 (1 - x^2)^{n+1} dx = \frac{1}{2(n+1)} I_{n+1}$

$$\therefore I_{n+1} = I_n - \frac{1}{2(n+1)} I_{n+1}, \quad \therefore I_{n+1} = \frac{2(n+1)}{2n+3} I_n.$$

iii) $I_n = \frac{2n}{2n+1} I_{n-1}, I_{n-1} = \frac{2(n-1)}{2n-1} I_{n-2}, \dots, I_2 = \frac{2 \times 2}{2 \times 2 + 1} I_1$
 $I_n = \frac{2n}{2n+1} \times \frac{2(n-1)}{2n-1} \times \dots \times \frac{2 \times 2}{2 \times 2 + 1} I_1$

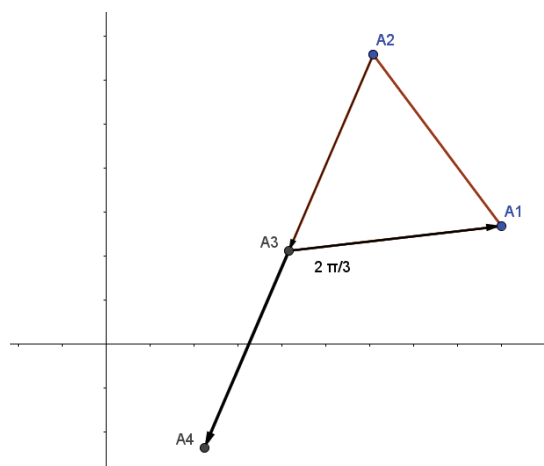
$$I_n = \frac{2n}{2n+1} \times \frac{2n}{2n} \times \frac{2(n-1)}{2n-1} \times \frac{2(n-1)}{2(n-1)} \times \dots \times \frac{2(2)}{2(2)} \times \frac{2 \times 2}{2 \times 2 + 1} I_1$$

$$= \frac{(2 \times 2 \times \dots \times 2)^2 \times (n \times (n-1) \times \dots \times 2 \times 1)^2}{(2n+1) \times (2n) \times (2n-1) \dots 5 \times 4 \times 3 \times 2 \times 1} = \frac{2^{2n} \times n!^2}{(2n+1)!}$$

Question 15 a)

i) $\overrightarrow{A_3 A_4} = \overrightarrow{A_2 A_3}$
 $\overrightarrow{A_3 A_1} = (\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}) \overrightarrow{A_3 A_4}$
 $= (\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}) \overrightarrow{A_2 A_3}$
 $= \omega \overrightarrow{A_2 A_3}$

ii) The triangle $A_1 A_2 A_3$ is inscribed in a circle.
Let O be the centre and r its radius.
 $OA_1 = OA_2 = OA_3 = r.$



Let z_0 the complex number corresponding to O. Since z_1, z_2, z_3 are the complex numbers corresponding to A_1, A_2, A_3 respectively.

Now

$$\overrightarrow{OA_3} = \omega \overrightarrow{OA_2}$$

$$\overrightarrow{OA_1} = \omega \overrightarrow{OA_3}$$

$$\overrightarrow{OA_2} = \omega \overrightarrow{OA_1}$$

$$\therefore z_3 - z_0 = \omega(z_2 - z_0) \quad (1)$$

$$z_1 - z_0 = \omega(z_3 - z_0) \quad (2)$$

$$z_2 - z_0 = \omega(z_1 - z_0) \quad (3)$$

Add the three equations $\therefore$

$$z_1 + z_2 + z_3 - 3z_0 = \omega(z_1 + z_2 + z_3) - 3\omega z_0$$

$$(1 - \omega)(z_1 + z_2 + z_3) = 3(1 - \omega)z_0$$

$$z_0 = \frac{1}{3}(z_1 + z_2 + z_3)$$

In triangle OA_1A_2 , applying the cosine rule:

$$A_2A_1^2 = OA_1^2 + OA_2^2 - 2 \times OA_1 \times OA_2 \cos \frac{2\pi}{3}$$

$$|z_1 - z_2|^2 = r^2 + r^2 - 2 \times r \times r \times \cos \frac{2\pi}{3} = 3r^2$$

$$r = \frac{1}{\sqrt{3}}|z_1 - z_2|.$$

iii) ω is the complex cube root of unity $\therefore \omega^3 = 1, \omega^3 - 1 = 0, (\omega - 1)(\omega^2 + \omega + 1) = 0$

$$\omega^2 + \omega + 1 = 0 \quad (*)$$

Using (i) $z_1 - z_3 = \omega(z_3 - z_2)$

$$z_1 + \omega z_2 - (1 + \omega)z_3 = 0, \text{ but from } (*) \omega + 1 = -\omega^2$$

$$\therefore z_1 + \omega z_2 + \omega^2 z_3 = 0.$$

iv) Using (iii) $z_1 = -(\omega z_2 + \omega^2 z_3)$, also $z_2 = -(\omega z_3 + \omega^2 z_1)$, $z_3 = -(\omega z_1 + \omega^2 z_2)$.

$$z_1^2 + z_2^2 + z_3^2 = [-(\omega z_2 + \omega^2 z_3)]^2 + [-(\omega z_3 + \omega^2 z_1)]^2 + [-(\omega z_1 + \omega^2 z_2)]^2$$

$$= \omega^2 z_2^2 + 2\omega \omega^2 z_2 z_3 + \omega^4 z_3^2 + \omega^2 z_3^2 + 2\omega \omega^2 z_1 z_3 + \omega^4 z_1^2 + \omega^2 z_1^2 + 2\omega \omega^2 z_1 z_2 + \omega^4 z_2^2$$

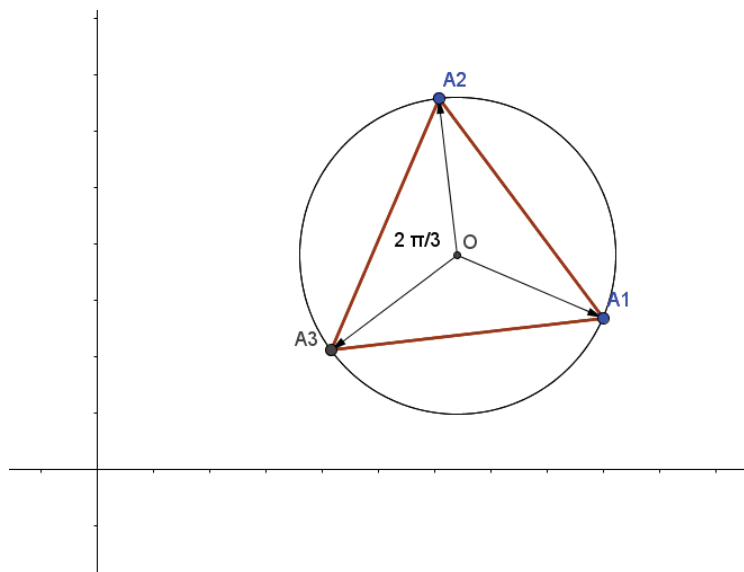
$$= 2\omega^3(z_1 z_2 + z_2 z_3 + z_1 z_3) + (\omega^2 + \omega^4)(z_1^2 + z_2^2 + z_3^2)$$

But $\omega^3 = 1, \omega^4 = \omega$ and $\omega^2 + \omega^4 = -1$.

$$z_1^2 + z_2^2 + z_3^2 = 2(z_1 z_2 + z_2 z_3 + z_1 z_3) - (z_1^2 + z_2^2 + z_3^2)$$

$$2(z_1^2 + z_2^2 + z_3^2) = 2(z_1 z_2 + z_2 z_3 + z_1 z_3)$$

$$z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_1 z_3.$$



b) i) Prove by mathematical induction that $y_n = \frac{2}{3^{n-1}} x_n$ $n \geq 0$.

$$y_0 = \frac{2}{3^{0-1}} x_0 = 6x_0 = 6 \times \frac{1}{2} [1 + 1] = 6, \text{ true for } n = 0.$$

$$y_1 = \frac{2}{3^{1-1}} x_1 = 2x_1 = 2 \times \frac{1}{2} [1 + i\sqrt{2} + 1 - i\sqrt{2}] = 2, \text{ true for } n = 1.$$

Assume it is true for $n = k$, i.e. $y_k = \frac{2}{3^{k-1}} x_k$ $k \geq 0$ (**) Induction Hypothesis

And prove it true for $n = k + 1$.

$$\text{i.e. } y_{k+1} = \frac{2}{3^k} x_{k+1}$$

Using the recurrence relation of y , $\therefore 3y_{k+1} = 2y_k - y_{k-1} = 2 \times \frac{2}{3^{k-1}} x_k - \frac{2}{3^{k-2}} x_{k-1}$

$$\begin{aligned} &= \frac{1}{3^{k-1}} [2(1 + i\sqrt{2})^k + 2(1 - i\sqrt{2})^k - 3(1 + i\sqrt{2})^{k-1} - 3(1 - i\sqrt{2})^{k-1}] \\ &= \frac{1}{3^{k-1}} [(1 + i\sqrt{2})^{k-1} \{2(1 + i\sqrt{2}) - 3\} + (1 - i\sqrt{2})^{k-1} \{2(1 - i\sqrt{2}) - 3\}] \\ &= \frac{1}{3^{k-1}} [(1 + i\sqrt{2})^{k-1} (-1 + 2i\sqrt{2}) + (1 - i\sqrt{2})^{k-1} (-1 - 2i\sqrt{2})] \\ &= \frac{1}{3^{k-1}} [(1 + i\sqrt{2})^{k-1} (1 + i\sqrt{2})^2 + (1 - i\sqrt{2})^{k-1} (1 - i\sqrt{2})^2] \\ &= \frac{1}{3^{k-1}} [(1 + i\sqrt{2})^{k+1} + (1 - i\sqrt{2})^{k+1}] \\ &= \frac{2}{3^{k-1}} \times \frac{1}{2} [(1 + i\sqrt{2})^{k+1} + (1 - i\sqrt{2})^{k+1}] \\ &3y_{k+1} = \frac{2}{3^{k-1}} x_{k+1} \\ &y_{k+1} = \frac{2}{3^k} x_{k+1} \end{aligned}$$

Hence by mathematical induction it is true for all $n \geq 0$.

$$\begin{aligned} \text{ii) } y_n &= \frac{2}{3^{n-1}} x_n = \frac{2}{3^{n-1}} \times \frac{1}{2} [(1 + i\sqrt{2})^n + (1 - i\sqrt{2})^n] \\ &= \frac{1}{3^{n-1}} [(1 + i\sqrt{2})^n + (1 - i\sqrt{2})^n] \\ &= 3 \left[\left(\frac{1 + i\sqrt{2}}{3} \right)^n + \left(\frac{1 - i\sqrt{2}}{3} \right)^n \right] \\ &= 3 \left[\left(\frac{1 + i\sqrt{2}}{3} \times \frac{1 - i\sqrt{2}}{1 - i\sqrt{2}} \right)^n + \left(\frac{1 - i\sqrt{2}}{3} \times \frac{1 + i\sqrt{2}}{1 + i\sqrt{2}} \right)^n \right] \\ &= 3 \left[\left(\frac{1}{1 - i\sqrt{2}} \right)^n + \left(\frac{1}{1 + i\sqrt{2}} \right)^n \right] \end{aligned}$$

Since $(1 + i\sqrt{2}) \times (1 - i\sqrt{2}) = 3$.

Question 16

a) i) $a + b \geq 2\sqrt{ab}$

$$a^2 + b^2 \geq 2ab$$

$$a^2 + c^2 \geq 2ac$$

$$b^2 + c^2 \geq 2bc$$

$$a^2 + b^2 + a^2 + c^2 + b^2 + c^2 \geq 2ab + 2ac + 2bc$$

$$2(a^2 + b^2 + c^2) \geq 2(ab + ac + bc)$$

$$a^2 + b^2 + c^2 \geq ab + ac + bc$$

ii) $a + b + c \geq 3\sqrt[3]{abc}$

$$\frac{a^3}{b-c} + \frac{b^3}{c-a} + (b-c)(c-a) \geq 3ab$$

$$\frac{a^3}{b-c} + \frac{c^3}{a-b} + (b-c)(a-b) \geq 3ac$$

$$\frac{b^3}{c-a} + \frac{c^3}{a-b} + (c-a)(a-b) \geq 3bc$$

$$2\left[\frac{a^3}{b-c} + \frac{b^3}{c-a} + \frac{c^3}{a-b}\right] + (b-c)(c-a) + (b-c)(a-b) + (c-a)(a-b) \geq 3(ab + ac + bc)$$

Now $(b-c)(c-a) + (b-c)(a-b) + (c-a)(a-b) = bc - ab - c^2 + ac + ab - b^2 - ac + bc + ac - bc - a^2 + ab = ab + ac + bc - (a^2 + b^2 + c^2)$

$$2\left[\frac{a^3}{b-c} + \frac{b^3}{c-a} + \frac{c^3}{a-b}\right] + ab + ac + bc - (a^2 + b^2 + c^2) \geq 3(ab + ac + bc)$$

$$2\left[\frac{a^3}{b-c} + \frac{b^3}{c-a} + \frac{c^3}{a-b}\right] \geq -(ab + ac + bc) + (a^2 + b^2 + c^2) + 3(ab + ac + bc) \geq 3(ab + ac + bc)$$

$$\frac{a^3}{b-c} + \frac{b^3}{c-a} + \frac{c^3}{a-b} \geq \frac{3}{2}(ab + ac + bc)$$

b) i) Let $f(x) = x - \ln(1+x)$

$$f'(x) = 1 - \frac{1}{1+x} = \frac{x}{1+x} \geq 0, \text{ for } x \geq 0$$

$f(x)$ is increasing function.

$$f'(x) = 0 \therefore x = 0$$

$$f''(x) = \frac{1}{(1+x)^2} > 0 \therefore (0, f(0)) \text{ is a minimum.}$$

$$f(0) = 0 \text{ which is minimum and } f(x) \text{ is increasing for } x > 0$$

$$f(x) > 0 \text{ for } x > 0.$$

ii) $x_n = \left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{3^2}\right) \dots \left(1 + \frac{1}{3^n}\right)$

$$x_{n+1} = \left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{3^2}\right) \dots \left(1 + \frac{1}{3^n}\right)\left(1 + \frac{1}{3^{n+1}}\right)$$

$$\frac{x_{n+1}}{x_n} = \frac{\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{3^2}\right) \dots \left(1 + \frac{1}{3^n}\right)\left(1 + \frac{1}{3^{n+1}}\right)}{\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{3^2}\right) \dots \left(1 + \frac{1}{3^n}\right)} = 1 + \frac{1}{3^{n+1}} > 1$$

$$x_{n+1} > x_n.$$

$$\text{iii) } \frac{x_{n+1}}{x_n} = 1 + \frac{1}{3^{n+1}}, \text{ so } \frac{x_{k+1} - x_k}{x_k} = 1 + \frac{1}{3^{k+1}} - 1 = \frac{1}{3^{k+1}}$$

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{x_{k+1} - x_k}{x_k} = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{3^{k+1}} = \frac{\frac{1}{3^2}}{1 - \frac{1}{3}} = \frac{1}{6}$$

$$\text{iv) } x_n = \left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{3^2}\right) \dots \left(1 + \frac{1}{3^n}\right)$$

$$\ln x_n = \ln\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{3^2}\right) \dots \left(1 + \frac{1}{3^n}\right) = \ln\left(1 + \frac{1}{3}\right) + \ln\left(1 + \frac{1}{3^2}\right) + \dots + \ln\left(1 + \frac{1}{3^n}\right)$$

Using b)(ii)

$$\ln x_n < \frac{1}{3} + \frac{1}{3^2} + \dots + \frac{1}{3^n} = \sum_{k=1}^n \frac{1}{3^k}$$

$$\text{v) } \ln x_n < \sum_{k=1}^n \frac{1}{3^k} < \sum_{k=1}^{\infty} \frac{1}{3^k} = \frac{\frac{1}{3}}{1 - \frac{1}{3}} = \frac{1}{2}$$

$$e^{\ln x_n} < e^{\frac{1}{2}}$$

$$x_n < \sqrt{e}.$$

The End.